

KPP-01 Buoyancy-Driven Pneumatic DEMO SIMULATOR - Governing Physics

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Plant / code: D:\PNEUMATIC project v1 (hmi\_demo)  
Live HMI: http://127.0.0.1:4173/  
Grounding snapshot: GET http://127.0.0.1:4173/api/grok/snapshot  
capturedAt: 2026-09-18T19:23:38.601Z  
Fidelity: engineering | demoAssist live: 0 N|m | Plant status: ready

> Every numeric claim below is taken from that snapshot's live tags and/or designBasis / designBasisAdvice / designBasisWarnings. No invented plant constants.

Abstract

KPP-01's HMI simulator integrates open-cycle pneumatic injection, hydrostatic floater gas state, buoyant torque on an endless chain, rotational rigid-body dynamics, geared generation, and an energy ledger each tick. Design-basis net plant power is 213.257 kW (compressorPowerKW = 15.843, designElectricalKW = 2.586). The plant is air-supply limited near 2.59 kW electrical - not the 8 kW generator nameplate. With receiver cut-out 8 bara and dock injection need 1.90 bara, about 11.0 kW is throttled at the regulator (~69% of design compressor shaft power). In engineering fidelity, demoAssist == 0 and must never be labelled buoyancy.

Keywords: buoyancy; open-cycle pneumatics; Boyle/polytropic floater gas; hydrostatic injection; throttle loss; energy balance; KPP-01 simulator

1. What the plant is

KPP-01 is an open-cycle buoyancy pneumatic demonstration:

- 1. Compressor fills a receiver.
- 2. Regulator drops pressure to a header setpoint.
- 3. At the dock (bottom), air is injected into floaters against hydrostatic head.
- 4. Air-filled floaters ascend; empty/vented floaters descend.
- 5. Net buoyant torque turns the tower -> gearbox -> generator.
- 6. At the top, air is vented to atmosphere (open cycle). Compression work in that air is largely rejected.

Simulator modules (code): FloaterPhysics.js, TowerDynamics.js, Config.js (pressureAtDepth, designBasis builders), PlantSnapshot.js (API exposure).

2. Nomenclature

| Symbol                                                        | Meaning                                                 | SI / plant unit |
|---------------------------------------------------------------|---------------------------------------------------------|-----------------|
| $P_{\mathrm{atm}}$                                            | Surface absolute pressure                               | bara            |
| $h$                                                           | Depth below free surface                                | m               |
| $\rho$                                                        | Water density (WATER_DENSITY)                           | 1000 kg/m³      |
| $g$                                                           | Gravity                                                 | m/s²            |
| $P_{\mathrm{dock}}$                                           | Absolute injection pressure at dock                     | bara            |
| $V$                                                           | Displaced / gas volume of a floater                     | m³              |
| $m$                                                           | Floater structure mass (plus entrained terms in design) | kg              |
| $F_b$                                                         | Net buoyant force                                       | N               |
| $r$                                                           | Sprocket radius (sprocketRadiusM)                       | m               |
| $\tau$                                                        | Torque                                                  | N m             |
| $\omega$                                                      | Angular velocity                                        | rad/s           |
| $I$                                                           | Tower rotational inertia                                | kg m²           |
| $\dot{V}_{\mathrm{FAD}}$                                      | Compressor free-air delivery                            | m³/min          |
| $P_{\mathrm{gen}}$ , $P_{\mathrm{comp}}$ , $P_{\mathrm{net}}$ | Electrical out, compressor in, net                      | kW              |
| $\tau_{\mathrm{assist}}$                                      | demoAssist torque (not buoyancy)                        | N m             |

3. Hydrostatics at the dock

3.1 Governing equation

Fresh-water hydrostatic gradient in the simulator is 0.1 bar/m (BAR\_PER\_METRE = 0.1 in FloaterPhysics.js):

$$P(h) = P_{\mathrm{atm}} + (0.1\,\mathrm{bar/m})h$$

equivalently

$$P_{\mathrm{dock}} = P_{\mathrm{atm}} + \rho g h$$

\quad\text{with}\quad

$$\frac{\rho g}{10^5} \approx 0.0981 \, \mathrm{bar/m} \approx 0.1 \, \mathrm{bar/m}.$$

Code: ambientPressureAt(depthM, atmBara) and pressureAtDepth(depthM) in Config.js.

### 3.2 Snapshot / designBasis numbers

| Quantity                                                | Value                                       | Source      |
|---------------------------------------------------------|---------------------------------------------|-------------|
| designBasis.injectionPressureBara                       | 1.9                                         | designBasis |
| Live pneumatic.injection.requiredBara (ready/idle dock) | 1.0 (no active deep fill demand this frame) | live        |
| Implied dock depth at 1.9 bara if $P_{\mathrm{atm}}=1$  | $h=(1.9-1)/0.1=9$ m                         | derived     |

Physics truth for running fill: designBasis injection head is 1.90 bara.

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## 4. Boyle / polytropic free-air volume for floater fill

### 4.1 Isothermal (Boyle) mass-volume relation

Default gas law in FloaterPhysics.js is isothermal:

$$m = \rho_1 P, V$$

\quad\rightarrow\quad

$$V = \frac{m}{\rho_1 P}$$

with  $P$  in bara and  $\rho_1$  = air density at 1 bara.

Free-air equivalent volume to fill geometric volume  $V_f$  at dock pressure:

$$V_{\mathrm{free}} = V_f \cdot \frac{P_{\mathrm{dock}}}{P_{\mathrm{atm}}}$$

Example: if  $V_f = 0.16 \, \mathrm{m}^3$  and  $P_{\mathrm{dock}} = 1.9 \, \mathrm{bara}$ ,  $V_{\mathrm{free}} = 0.304 \, \mathrm{m}^3$  per fill (illustrative geometry; live inventory uses mass tags).

### 4.2 Polytropic extension

For exponent  $n \in [1, 1.4]$ :

$$V = V_{\mathrm{ref}} \left( \frac{P_{\mathrm{ref}}}{P} \right)^{1/n}$$

$n=1$  recovers Boyle;  $n>1$  reduces mean ascending volume (less buoyancy work per injected mass). Code: expansionVolume(...).

### 4.3 Live air inventory (this snapshot)

| Tag                             | Value                                              |
|---------------------------------|----------------------------------------------------|
| mechanical.floaters.airMassKg   | 2.156 kg                                           |
| pneumatic.accounting.injectedKg | 2.156 kg                                           |
| ventedKg / recoveredKg          | 0 / 0 (plant ready; no active lap vent this frame) |

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## 5. Buoyant force, torque, and shaft power

### 5.1 Force on one floater

$$F_b = (\rho V - m) g$$

Ascending (air volume  $V$ ): typically  $F_b > 0$  (up).

Descending (vented / water-filled): buoyancy shrinks; weight dominates.

Design builder in Config.js uses the same idea for mean ascending force:

$$F_{\mathrm{asc,mean}} = \bar{V}_{\mathrm{air}} \rho g - W_{\mathrm{structure}}$$

5.2 Torque sum on the sprocket

$$\tau_{\mathrm{buoyancy}} = \sum_i F_{b,i} r_i \\ \quad \quad \quad (r \approx \texttt{sprocketRadiusM})$$

Critical fidelity rule:

$$\tau_{\mathrm{driving}} = \tau_{\mathrm{buoyancy}} + \tau_{\mathrm{assist}}$$

with  $\tau_{\mathrm{assist}} = \texttt{demoAssist}$  - not buoyancy.

| Quantity                       | Value         | Source                             |
|--------------------------------|---------------|------------------------------------|
| Live torqueNm.buoyancy         | 10779.7 N m   | snapshot                           |
| Live torqueNm.demoAssist       | 0 N m         | snapshot                           |
| designBasis.designTorqueNm     | 10796.809 N m | designBasis                        |
| designBasis.demoAssistTorqueNm | 17841.538 N m | what-if only - not live eng torque |

5.3 Mechanical power

$$P_{\mathrm{mech}} = \tau \omega, \quad \omega = 2\pi \texttt{n}/60$$

Design check:

$$P_{\mathrm{mech,design}} = 10796.809 \times \frac{2\pi}{60} \approx 3.114 \text{ kW}$$

matches designBasis.designMechanicalKW = 3.114.

Live (ready/braked): towerRPM = 0, so  $P_{\mathrm{mech, live}} = 0$  this frame; buoyancy torque is held against brake (brake = 14500 N|m).

5.4 Rotational dynamics each tick (TowerDynamics.updateDynamics)

Newton's second law for rotation:

$$I \alpha = \tau_{\mathrm{net}}, \quad \alpha = \frac{\tau_{\mathrm{net}}}{I}$$

$$\omega \mapsto \omega + \alpha \Delta t$$

Resistive torques (friction, water drag, brake, generator load) oppose motion; they stop the shaft but do not reverse-drive it. Code zero-cross clamps prevent chatter.

Chain speed:

$$v_{\mathrm{chain}} = \omega \cdot r_{\mathrm{sprocket}}$$

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## 6. Pneumatic chain: receiver, regulator, FAD vs demand

6.1 Snapshot setpoints & sensors

| Quantity                | Value | Source |
|-------------------------|-------|--------|
| setpoints.receiverBara  | 8     | live   |
| setpoints.regulatorBara | 2.3   | live   |
| sensors.ptTankBara      | 5.175 | live   |

```
| sensors.ptHeaderBara | 2.306 | live |
| Compressor FAD | 3.2 m³/min | live + designBasis |
| designBasis.airDemandM3Min | 3.2 | designBasis |
| designBasis.compressorPowerKW | 15.843 | designBasis |
```

6.2 Why 8 bara vs ~1.9 dock wastes ~69%

Idealized isothermal compressor work to pressure  $\backslash(P)\backslash$  scales as  $\backslash(P\backslash\ln(P/P_0)\backslash)$  (or catalogue curve at cut-out). DesignBasis warning states explicitly:

> Receiver set to 8 bar but injection only needs 1.90 bara: 15.84 kW of compression where 4.89 kW would do. The surplus is throttled away at the regulator.

$$\eta_{\mathrm{throttle\,waste}} = \frac{15.843 - 4.89}{15.843} \approx 0.691 \quad \backslash;(69\% \backslash)$$

designBasisAdvice: ?About 11.0 kW is throttled away at the regulator.? Suggestion: cut-out 2.3 bar (dock + 0.4 bar).

Regulator drop:

$$\Delta P_{\mathrm{throttle}} = P_{\mathrm{receiver}} - P_{\mathrm{header}}$$

is irreversible for plant electrical output: shaft power spent raising pressure above dock need does not increase buoyant  $\backslash(V)\backslash$  at the dock.

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## 7. Open-cycle vent mass and energy

### 7.1 Mass balance tags

$$m_{\mathrm{injected}} = m_{\mathrm{vented}} + m_{\mathrm{held}} + m_{\mathrm{spilled}} - m_{\mathrm{recovered}} + \backslash\varepsilon \backslash$$

This snapshot (ready): injected 2.156 kg, vented 0, recovered 0, balanceErrorKg = 0.

During a running lap, design/intent is vent at top -> open cycle (recoveredKg remains 0 unless a recovery model is enabled).

### 7.2 Energy ledger (energyKWh)

```
| Channel | kWh (this snapshot window) |
|-----|-----|
| in | 0.0753 |
| out | 0 |
| compressionLoss | 0.0226 |
| throttle | 0 |
| spillVent | 0.0022 |
| assist | 0 |
| unexplainedFrac | 0.5178 (short/idle window - not a steady run) |
```

On a long engineering run, throttle and compression dominate in ? out, consistent with designBasis.

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## 8. Generator path: air-limited ~2.6 kW vs 8 kW nameplate

```
| Quantity | Value | Source |
|-----|-----|-----|
| electrical.generator.ratedKW | 8 | live |
| designBasis.designElectricalKW | 2.586 | designBasis |
| electrical.generator.loadRequestKW | 2.6 | live |
| Live delivered (ready) | 0 | tower stopped |
```

Advice: ?An 8 kW generator at this torque needs about 9.90 m³/min FAD? selected compressor delivers 3.20 m³/min, which supports 2.59 kW.?

$$P_{\mathrm{elec,air\text{-}limit}} \approx 2.59 \backslash,\mathrm{kW} \backslash \parallel 8 \backslash,\mathrm{kW} \backslash _{\mathrm{nameplate}}$$

Gear: fittedGearRatio = 108.9 (design recommended 108.911) - matches gen ~300 rpm at tower ~2.755 rpm when running.

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## 9. Net plant power

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$$P_{\mathrm{net}} = P_{\mathrm{gen}} - P_{\mathrm{comp}}$$

\\

$$| \text{Basis} | \backslash (P_{\mathrm{gen}}) | \backslash (P_{\mathrm{comp}}) | \backslash (P_{\mathrm{net}}) |$$

$$| \text{-----} | \text{-----} | \text{-----} | \text{-----} |$$

$$| \text{designBasis} | 2.586 \text{ kW} | 15.843 \text{ kW} | ?13.257 \text{ kW} |$$

$$| \text{live plantPower (this frame)} | 0 | 12.27 \text{ kW} | ?12.271 \text{ kW} |$$

designBasisWarnings: ?Design-basis net plant power is -13.26 kW (2.59 kW generated vs 15.84 kW of compression).?

Interpretation: negative net is expected open-cycle physics with over-pressured receiver - not a sensor bug.

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## 10. demoAssist vs buoyancy (engineering fidelity)

| | Live engineering | designBasis field |

| -- | ----- | ----- |

| Buoyancy torque | 10779.7 N|m | designTorqueNm 10796.809 |

| demoAssist | 0 N|m | demoAssistTorqueNm 17841.538 (counterfactual) |

| energyKWh.assist | 0 | - |

| fidelityMode | engineering | - |

Rule: Never call demoAssist buoyancy. Engineering acceptance requires  $\tau_{\mathrm{assist}} \equiv 0$ .

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## 11. What the simulator is solving each tick

State vector (conceptual; tags in State.js):

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$\mathbf{x} = \text{big}\{$

$\omega, \theta,$

$m_i, V_i, \text{segment}_i, \dots, \}_{i=1}^N,$

$P_{\mathrm{tank}}, P_{\mathrm{header}},$

$\text{valve/compressor flags},$

$E_{\mathrm{ledger}},$

$\text{gen load}$

$\text{big}\}$

\\

One  $(\Delta t)$  step (engineering fidelity):

1. Hydrostatics / gas state - for each floater,  $(P_{\mathrm{amb}}(h) = P_{\mathrm{atm}} + 0.1, h)$ ; update mass/volume via Boyle/polytropic (FloaterPhysics).
2. Dock injection - if docked and header above requiredBara, add air mass; demand accumulates toward FAD limit.
3. Vent / spill - at top segment, open-cycle mass leave; accounting tags update.
4. Forces -> torques -  $(F_{b,i} = (\rho V_i - m_i)g)$ ;  $(\tau_{\mathrm{buoy}} = \sum F_{b,i} r_i)$ ; add demoAssist only if enabled (0 in eng.).
5. Resistive torques - friction, bearing, water drag, brake, generator load.
6. Rigid-body update -  $(\alpha = \tau_{\mathrm{net}}/I)$ ; integrate  $(\omega, \theta)$ ; derive RPM & chain speed (TowerDynamics.updateDynamicics).
7. Pneumatic plant - compressor toward receiver setpoint; regulator toward header setpoint; throttle loss enters energy ledger.
8. Electrical - gearbox ratio maps  $(\omega \rightarrow \omega_g)$ ; delivered kW  $\leq$  air-limited / torque-limited available power; never silently equal to 8 kW nameplate.
9. Energy ledger - integrate compressor in, gen out, compression/throttle/vent/drag/friction/brake/gearbox/converter/assist channels.
10. Snapshot API - PlantSnapshot publishes tags + recomputed designBasis block for GET /api/grok/snapshot.

Loop geometry this basis: loopLengthM = 21.705, floaterPitchM = 0.748, N = 29 floaters, lapTimeSec = 86.989 at design RPM, fillsPerMin = 20.003.

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## 12. Equation index (quick)

1.  $(P_{\mathrm{dock}} = P_{\mathrm{atm}} + \rho g h)$  (0.1 bar/m in code)
2.  $(V_{\mathrm{free}} = V_f P_{\mathrm{dock}} / P_{\mathrm{atm}})$  (Boyle fill)
3.  $(F_b = (\rho V - m)g)$ ;  $(\tau = \sum F_b r)$ ;  $(P = \tau \omega)$
4.  $(\alpha = \tau_{\mathrm{net}}/I)$ ;  $(\omega \rightarrow \omega + \alpha \Delta t)$
5. Throttle waste  $(\approx (15.843 - 4.89) / 15.843 \approx 69\%)$  at receiver 8 vs dock 1.9

6.  $(P_{\text{net}} = P_{\text{gen}} - P_{\text{comp}})$  -> designBasis ?13.257 kW  
7.  $(\tau_{\text{assist}} \leq 0)$  in engineering (!= buoyancy)

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### 13. Conclusions

1. The simulator is a coupled hydrostatic-pneumatic-mechanical-electrical integrator, not a cartoon.
2. Design-basis net ?13.257 kW with air-limited electrical 2.586 kW is the honest stock plant.
3. ~69% of design compressor power is regulator throttle from receiver 8 bara vs dock 1.9 bara.
4. 8 kW is nameplate ceiling, not delivered physics.
5. demoAssist = 0 in engineering; designBasis assist torque is a what-if only.
6. Net  $\geq 0$  requires pressure matching and vent recovery / closed cycle - not a bigger generator sticker.

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### 14. Data citation

- Snapshot: capturedAt = 2026-09-18T19:23:38.601Z via GET <http://127.0.0.1:4173/api/grok/snapshot>
- Blocks used: plant, pneumatic, mechanical.torqueNm, electrical, energyKWh, designBasis, designBasisAdvice, designBasisWarnings
- Code references: hmi\_demo/js/utis/FloaterPhysics.js, Config.js (pressureAtDepth, WATER\_DENSITY), models/TowerDynamics.js (updateDynamics)
- Author actions: read-only; no plant START/STOP/setpoint writes for this document.
- End -